

A novel supercapacitor design that offers multiple advantages

Researchers from Chung-Ang University have fabricated a structure comprising multilayer electrodes vertically integrated within a single sheet of paper. The novel design overcomes the problems associated with stacked sheets while retaining the inherent advantages of a paper based substrate. The supercapacitor design showed a low electrical resistance, high foldability, and good mechanical strength. Manganese dioxide enhanced its active surface area, which further boosted the electrochemical performance. Additionally, the supercapacitor demonstrated high energy storage with maximum areal energy and power densities of $13.73\mu\text{W}\cdot\text{hr}\cdot\text{cm}^2$ and $1.6\text{mW}\cdot\text{cm}^2$, respectively. Moreover, it retained its storage capacity even after undergoing 6,000 charge-discharge cycles. Researchers say that their method of fabrication circumvents most fabrication challenges related to two-dimensional energy storage sheets.

Now an artificial throat with 99% speech accuracy developed

Researchers at Beijing's Tsinghua University have developed an artificial throat capable of recognising speech elements and generate speech at an accuracy rate of 99%. They designed a wafer-like tool one centimetre square that can easily be affixed to the throat and allow barely audible sounds, or even whispers, to be converted into speech at normal volume. Researchers refer to this device as a "graphene-based intelligent, wearable artificial throat." The 25-micrometre deep device may be applied to one's throat with a simple adhesive. Tiny wires connect to a microcontroller powered by a coin-sized battery. This artificial throat is capable of recognising speech elements such as phonemes, tones, and words at an accuracy rate of 99%. It relies on an artificial intelligence model to interpret captured sounds and bodily vibrations to generate vocalisations.

Algos for robots to adapt to the surroundings and learn being created

Columbia Engineering and Toyota Research Institute computer scientists are looking into psychology, physics, and geometry to create algorithms so that robots can adapt to their surroundings and learn how to do things independently. Researchers taught neural networks the basic physical concepts that come naturally to adults and children. The key idea is to train the computer to anticipate what the scene would look like in the future. Researchers modified the Iterative Residual

Policy (IRP) algorithm, which showed sub-inch accuracy by demonstrating its strong generalising capabilities. They developed a new approach to manipulating them by using actively blown air. The self-supervised learning framework

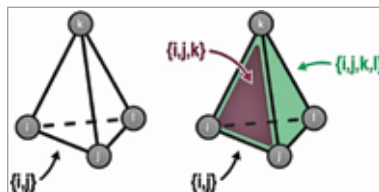


The self-supervised learning framework Columbia Engineers call DextAIRity learns to effectively perform a target task through a sequence of grasping or air based blowing actions. Using visual feedback, the system uses a closed-loop formulation that continuously adjusts its blowing direction (Credit: Zhenjia Xu/Columbia Engineering)

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Insights from biology help improve computer model of memory

Researchers at the Okinawa Institute of Science and Technology (OIST) have made improvements to a widely utilised computer model of memory, known as a Hopfield network, by incorporating insights from biology. The network is 'trained' to encode these patterns, then researchers test its



In the classical Hopfield network (left), each neuron (i, j, k, l) is connected to the others in a pairwise manner. In the modified network made by Burns and Professor Fukai, sets of three or more neurons can connect simultaneously (Credit: Thomas Burns, OIST)

memory by presenting a series of the blurry or incomplete patterns and seeing if the network can recognise them as one it already knows. Researchers created a modified Hopfield network in which not just pairs of neurons but sets of three, four,

or more neurons could link up too, such as might occur in the brain through astrocytes and dendritic trees. The new network allowed these so-called 'set-wise' connections; overall it contained the same total number of connections as before. Researchers found that a network containing a mix of both pairwise and setwise connections performed best and retained the highest number of memories.